

OPERATING SYSTEMS

UE25CS242A

Scheduling Algorithms

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Slides Credits for all the PPTs of this course



- The slides/diagrams in this course are an **adaptation**, **combination**, and **enhancement** of material from the following resources:

1. "Operating System Concepts", Abraham Silberschatz, Peter Baer Galvin, Greg Gagne 10th Edition, John Wiley & Sons, Global Edition, 2023.
2. "Advanced Programming in the Unix Environment", Richard Stevens and Stephen A Rago, Pearson, 3rd edition, 2017.
3. Some conceptual text and diagram from "Operating Systems, Internals and Design Principles", William Stallings, 9th Edition, Pearson, 2018.

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Priority and Round Robin Scheduling

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Priority Scheduling

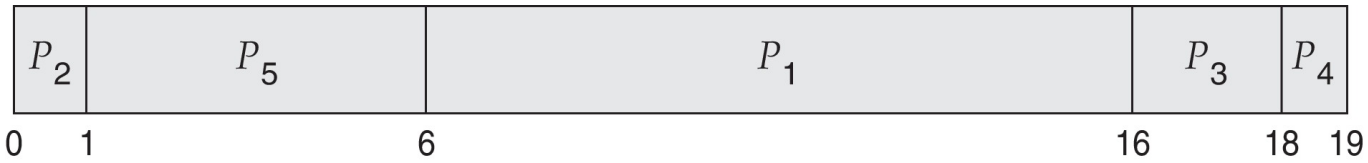


- A priority number (integer) is associated with each process
- The CPU is allocated to the process with the highest priority (smallest integer ≡ highest priority)
  - Preemptive
  - Nonpreemptive
- SJF is priority scheduling where priority is the inverse of predicted next CPU burst time
- Problem ≡ **Starvation** – low priority processes may never execute
- Solution ≡ **Aging** – as time progresses increase the priority of the process

Example of Non preemptive Priority Scheduling

Process	Burst Time	Priority
P_1	10	3
P_2	1	1
P_3	2	4
P_4	1	5
P_5	5	2

- Non preemptive Priority Scheduling Gantt chart



- Average waiting time = (6 + 0 + 16 + 18 + 1) / 5 = 41/5 = 8.2

Example of Preemptive Priority Scheduling

Process	Arrival Time	Burst Time	Priority
P_1	0	4	2
P_2	1	3	3
P3	2	1	4
P_4	3	5	5
P_5	4	2	5

- Preemptive Priority Scheduling Gantt chart
- Higher number represents higher priority



- Average waiting time = (11 + 8 + 0 + 0 + 4) / 5 = 23/5 = 4.6

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Round-Robin Scheduling



- Round-robin (RR) scheduling algorithm is designed especially for timesharing systems.
- It is similar to FCFS scheduling, but preemption is added to enable the system to switch between processes
- A small unit of time, called a time quantum or time slice, is defined.
- A time quantum is generally from 10 to 100 milliseconds in length
- The ready queue is treated as a circular queue
- The CPU scheduler goes around the ready queue, allocating the CPU to each process for a time interval of up to 1 time quantum

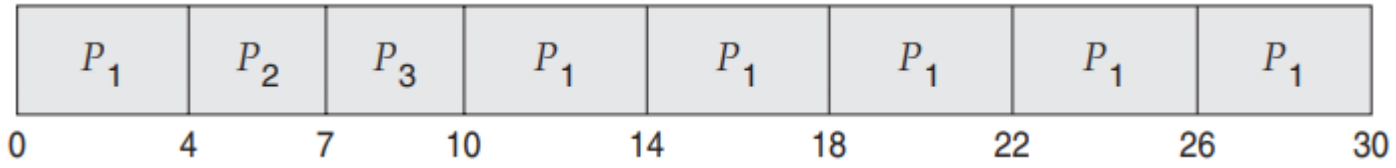
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Round-Robin Scheduling Example

- Consider the following set of processes that arrive at time 0, with the length of the CPU burst given in milliseconds:

Process	Burst Time
P_1	24
P_2	3
P_3	3

- RR scheduling Gantt chart using a time quantum of 4 milliseconds



  - P1 waits for 6 milliseconds (10 - 4)
  - P2 waits for 4 milliseconds
  - P3 waits for 7 milliseconds.

- The average waiting time = (6 + 4 + 7)/3 = 5.66 milliseconds

Round-Robin Scheduling Performance



- If there are n processes in the ready queue and the time quantum is q, then each process gets 1/n of the CPU time in chunks of at most q time units.
- Each process must wait no longer than (n − 1) × q time units until its next time quantum.
  - For example, with five processes and a time quantum of 20 milliseconds, each process will get up to 20 milliseconds every 100 milliseconds.
- Performance of the RR algorithm depends heavily on the size of the time quantum
- If the time quantum is extremely large, the RR policy is the same as the FCFS policy
- If the time quantum is extremely small, the RR approach can result in a large number of context switches



THANK YOU

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